Paris, France
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Sébastien **Gradit**

COMPUTATIONAL BIOLOGIST

sebgra
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sebastien-gradit

Bioengineer turned PhD Specialist with 5+ years in Bioinformatics, leveraging a high-level Machine Learning layer to build robust multi-omics platforms. I work at the intersection of AI and Genomics to transform complex biological data into scalable, actionable systems for precision therapeutics.

Skills

Core Programming	**Python**, **C++**, **R**, **High Performance Computing** [*Slurm*], **Cloud** [*AWS*]
Data Science	**Machine Learning** [*Keras*, *TensorFlow*, *PyTorch*, *SciKit-Learn*], **SQL**, **Interactive Dashboards** [*Streamlit, Shiny*]
Data Management	**CI/CD** [*GitHub Actions, Snakemake*], **Containerization** [*Docker, Singularity*], **Versioning** [*Git, GitHub*]
Bioinformatics	**Multi-Omics Integration**, **Automation of NGS**, **High-Throughput Analysis** [*RNA-Seq, Hi-C, Single-cell*]
Languages	**French** [*Native*], **English** [*Fluent: TOEIC 955/990*]

Experience

UFR Simone Veil — (UVSQ) *Paris, France*

DATA SCIENTIST - BIOINFORMATICIAN *2026 – Present*

- Optimizing advanced signal processing and normalization workflows for high-dimensional metabolomics data to ensure robust upstream data quality for machine learning models.
- Developing and benchmarking supervised machine learning frameworks and feature selection pipelines to discover clinical biomarkers for corticoid treatment response in severe sepsis cohorts.
- Implementing reproducible R/Python pipelines to analyze longitudinal clinical study data, translating complex metabolic profiles into actionable theranostic insights.

Pasteur Institute *Paris, France*

BIOINFORMATICIAN RESEARCHER | BIOINFORMATICS AND MACHINE LEARNING *2021 – 2025*

- Pioneered ”Hicberg” an ML-driven Python framework that recovers 100% of generated sequencing data, achieving a 30% information gain in repetitive genomic regions compared to existing methods.
- Engineered a platform-agnostic architecture with 95% validated fidelity against ground-truth, ensuring high-fidelity reconstruction across genomic data types.
- Unlocked previously ”unexploitable” therapeutic targets, providing a first-of-its-kind roadmap for precision medicine R&D.
- Architected industrial-grade R&D infrastructure (Snakemake, Docker, Slurm) to automate high-throughput processing on HPC clusters with 100% reproducibility.

Siemens HealthCare - LIPADE *Paris, France*

BIOMEDICAL ENGINEER *2020*

- Engineered a high-performance C++ feature extraction engine that automated bone lesion classification in xSPECT imaging, accelerating diagnostic workflows by 25% and eliminating manual contouring subjectivity for radiologists.
- Engineered an embedded ML pipeline for high-throughput clinical volume segmentation, enabling real-time, objective ROI classification and decision support.

Danone Nutricia Research *Saclay, France*

BIOMEDICAL ENGINEER *2019*

- Optimized clinical trial designs and pre-results analysis using ML-driven bioequivalence simulations and automated pipelines, accelerating insights by 50% and reducing study costs through optimal sample-size identification.

Education

PhD in Bioinformatics, PASTEUR INSTITUTE *Paris, France* *2021 - 2025*

M.Sc. in Image Processing and Artificial Intelligence, TÉLÉCOM PARIS - SORBONNE UNIVERSITY *Paris, France* *2019 - 2020*

M.Sc. in Biomedical Engineering (Engineer Diploma.), INSITUT SUPÉRIEUR DES BIOSCIENCES - ESIEE *Paris, France* *2016 - 2019*

Soft Skills

Soft Skills

- Analytical Thinking & Complex Problem-Solving
- Adaptability, Agility & Continuous Learning
- Creative Thinking & Innovation
- Effective Communication & Multidisciplinary Collaboration

Honors & Awards

2026 **Special Prize**, Digital 4 Genomics Hackathon — *Genopole, Evry*

Special Jury Prize: "Best Hackathoner" for sustained excellence across editions. Developed a machine learning framework for in silico virtual cell crafting utilizing Perturb-seq data to model cellular perturbations.

2025 **1st Place**, Read Write Grow - Biocatalyst Foundation — *Europe, Remote*

Classification of protein sequence safety through LLM and AI for Synthetic Biology designs

2024 **INCEPTION Program Grant**, Fellowship — *Paris, France*

Awarded a highly competitive grant covering the final three months of the PhD program, demonstrating recognition of research merit and strategic value to the field.

2022 **1st Place**, Digital 4 Genomics Hackathon — *Genopole, Evry*

Certificates

2023 **TensorFlow Advanced Specialization**, DeepLearning.AI

2023 **Deep Learning Specialization**, DeepLearning.AI

2023 **Machine Learning Specialization**, Stanford - DeepLearning.AI

Publications

Hicberg: Reconstruction of Contact Signals from Repeated Elements, **BiorXiv** - *In prep.*

Gradit S., *Ortion S.*, *Larrous P.*, *Delouis M.*, *Koszul R.*, *Cournac A.* - biorXiv doi: 10.1101/2025.06.20.660295

Selected scientific communications

2024 **JOBIM 2024, Toulouse**, Poster | Prediction of Omics Signal from Repeated Elements

2024 **JeBiF@JOBIM 2024, Toulouse**, Workshop | Good Practices in Bioinformatics

2023 **International Congress for Transposable Elements, St-Malo**, Poster | Prediction of Omics Signal from Repeated Elements

2023 **31st Intelligence Systems for Molecular Biology (ISMB), Lyon**, Poster and Talk | Statistical Inference of Repeated Elements Contacts in Hi-C maps

Extracurricular Activity

Jeunes BioInformaticiens de France (JeBiF) — *France*

President — *2022 - 2024*

- Led strategic initiatives to **structure and energize** France's emerging bioinformatics talent.
- Fostered **national and international collaborations** within the bioinformatics community.
- **Promoted and popularized** bioinformatics to public, private, and international stakeholders.
- Organized **community-building events** and disseminated information on bioinformatics training.